Dr. Anika Sharma

Scientist E — Atmospheric Instrumentation & Applied AI

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PROFESSIONAL SUMMARY

Electronics and avionics engineer with twelve years of airborne-systems integration and qualification testing, followed by ten years leading scientific instrumentation and edge-AI development in government service. Experienced in MIL-STD-1553 test rigs, UAV payload electronics, sensor telemetry, embedded GPU deployment and technical procurement.

AREAS OF EXPERTISE

- Avionics (MIL-STD-1553 / ARINC 429, Avionics integration)
- Scientific instrumentation (Environmental qualification testing, Sensor calibration, Radiosonde and upper-air systems)
- Embedded systems (Embedded C, ESP32 / STM32 microcontrollers)
- Edge AI (NVIDIA Jetson deployment)
- Drones & UAV systems (UAV payload integration)
- AI / machine learning (PyTorch, Deep learning)
- IoT (MQTT and Node-RED, InfluxDB and Grafana)
- Computer vision (Image super-resolution, Object detection)
- GPU computing (CUDA / GPU computing)
- Model optimisation (TensorRT and ONNX, Pruning and knowledge distillation)
- Software & tools (Python, Docker and Linux)
- Technical leadership (Team leadership)

PROFESSIONAL EXPERIENCE

Scientist E Jul 2016 – Present

Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory, New Delhi

Responsible for upper-air observation instruments, the division workshop and the laboratory's IoT and AI/ML activities.

- Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT.
- Led acceptance testing and commissioning of radiosonde ground-receiving systems at three observing stations, writing the test protocol and the defect-tracking procedure.
- Designed and deployed an IoT monitoring network for laboratory infrastructure (ESP32 sensor nodes, MQTT, Node-RED, InfluxDB and Grafana dashboards) covering temperature, humidity and power quality.
- Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules.
- Authored standard operating procedures for pre-flight checks and calibration of upper-air instruments used across the division.

Additional charge: **Workshop In-charge**, Upper Air Division Workshop Apr 2019 – Present

Additional charge of the division workshop supporting instrument repair, fabrication and calibration.

- Introduced a job-card and spares-inventory system for instrument repairs and calibration work.
- Supervise technicians and plan preventive maintenance for workshop test equipment.

Nodal officer: **Nodal Officer, IoT and AI/ML Initiatives** Jan 2021 – Present

Senior Design Engineer Jun 2004 – Jun 2016

Avionics Integration Group, Halcyon Avionics Design Bureau, Bengaluru

Integration and qualification testing of airborne electronics and UAV payloads.

- Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight trials.

Fictional sample profile — not a real person

- Integrated mission-computer and display units on MIL-STD-1553 avionics buses and built the automated test rig used for bench validation.
- Mentored graduate engineer trainees on avionics bus protocols and test documentation.

Graduate Engineer Aug 2002 – May 2004

Test Systems Group, Meridian Electronics Limited, Bengaluru

Built automated test equipment for avionics line-replaceable units.

- Developed LabVIEW test sequences for power-supply and interface boards of avionics units.

LEADERSHIP

Supervisor, M.Tech. interns in applied machine learning 2022 – Present

Kestrel Atmospheric Instrumentation Laboratory

Supervise summer and dissertation interns on edge-AI projects.

Chair, Technical Evaluation Committee for radiosonde procurement Apr 2023 – Feb 2024

Kestrel Atmospheric Instrumentation Laboratory

Chaired technical evaluation of bids for upper-air sounding systems.

- Evaluation report accepted by the competent authority.

EDUCATION

Ph.D., Electrical and Electronics Engineering 2021 – ongoing

Northfield Institute of Science and Technology

Thesis: Lightweight deep networks for real-time enhancement of aerial imagery

M.Tech., Avionics 2010

Coastal Institute of Aeronautical Engineering, Chennai

Grade: CGPA 8.6/10

Thesis: Fault-tolerant data acquisition for airborne test instrumentation

B.E., Electronics and Communication Engineering 2004

Northfield Institute of Science and Technology, Northfield University, Pune

Grade: First Class with Distinction

AWARDS & HONOURS

Best Paper Award 2023

Example Conference on Instrumentation (ECI 2023)

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